Vehicle Insurance & Claims Management

End-to-end digital platform — from instant quotes to direct bank payouts

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Traditional Insurance vs VehicleSure

Traditional Insurance

- Slow manual paperwork processes
- Disconnected teams and departments
- No real-time application tracking
- Manual claim assessment and errors
- Delayed payouts via cheque or NEFT
- No unified platform for all parties

VehicleSure Platform

- Instant online quote engine
- Role-based portals for all stakeholders
- Real-time status tracking and notifications
- Automated depreciation-based payout calculation
- Direct bank transfers via RazorpayX
- Single unified platform for all workflows

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Technology Stack

Frontend

Angular 21 · RxJS · Tailwind CSS

Backend

Java 17 · Spring Boot 3.3 · Spring Security (JWT)

Database

MySQL – relational schema, 11 entities

Premium Payment

Razorpay – order creation + HMAC signature verification

Claim Payout

RazorpayX – direct bank transfers to customer accounts

Email

n8n – webhook-triggered transactional emails

PDF Generation

OpenPDF – invoice and policy documents auto-generated

Excel Reports

Apache POI – premium and payout history exports

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System Roles

Customer

- Request quotes and buy policies
- Pay premiums via Razorpay
- Submit and track claims

Underwriter

- Review assigned applications
- Verify RC and invoice documents
- Approve or reject with reason

Claims Officer

- Review assigned claims
- Calculate depreciation payout
- Approve or reject claims

Admin

- Manage staff and policy plans
- Assign applications and claims
- Trigger RazorpayX payouts

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Workflow 1: Insurance Issuance

Quote → Application → Underwriting → Payment → Policy Activation

1

Get Quote

Customer enters vehicle details, gets instant premium options

2

Submit Application

Choose plan, upload RC and invoice documents

3

Assign Underwriter

Admin assigns to available underwriter based on workload

4

Review & Decide

Underwriter approves (creates policy) or rejects with reason

5

Pay Premium

Razorpay checkout, HMAC signature verified server-side

6

Activate Policy

PDF invoice and policy document generated, policy becomes active

7

Download Documents

Invoice and policy document available on dashboard

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Workflow 2: Claim Settlement

Phase 1: Submission & Assignment

01

File Claim

Customer submits with claim type, bank details, and up to 3 supporting documents

02

Assign Officer

Admin checks officer workload and assigns a Claims Officer

03

Review & Calculate

Officer assesses documents, computes depreciation payout

04

Approve / Reject

Set approved payout amount or reject with reason

Phase 2: Payout & Settlement

01

Trigger Payout

Admin processes RazorpayX bank transfer

02

Record & Notify

ClaimsPayment record saved; status changes to SETTLED

03

Receive Transfer

Amount credited to customer's bank account directly

04

Policy Update

Total-loss claims set policy to INACTIVE

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Key Features

1. **Smart Quote Engine** — IDV and risk scoring in real time
2. **Role-Based Dashboards** — Separate portals per stakeholder
3. **Document Upload** — RC and invoice upload with verification
4. **Razorpay Integration** — Secure order and HMAC signature flow
5. **Auto PDF Generation** — Invoice and policy documents on payment

1. **Multi-Type Policies** — Third Party, Comprehensive, Zero Dep
2. **Claims Filing Portal** — Self-service with 3-document support
3. **RazorpayX Payouts** — Direct bank transfer on approval
4. **In-App Notifications** — Real-time alerts at every step
5. **Excel Report Export** — Admin exports for audit trail

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Business Rules & Calculations

IDV Calculation

Formula: IDV = Ex-Showroom Price × (1 − Age × 10%)

Maximum depreciation capped at 50%

Example: 3-year car ₹8L → ₹8L × 0.70 = ₹5.6L

Example: 6-year car ₹5L → ₹5L × 0.50 = ₹2.5L (cap applies)

Premium Calculation

Formula: Base Premium × Risk Multiplier × IDV Multiplier

LOW risk (score ⩽1) → 1.0× | No increase

MEDIUM risk (score 2) → 1.2× | +20%

HIGH risk (score ⩾3) → 1.5× | +50%

Multi-year: 2 years = ×1.9 (5% off) | 3 years = ×2.8 (7% off)

Claim Payout Logic

THEFT → Payout = Full IDV; Policy closed

DAMAGE > 75% IDV → Total Loss; Payout = Full IDV

DAMAGE ⩽ 75% IDV → Comprehensive: Depreciated Bill | Zero-Dep: Full Bill (no depreciation)

THIRD PARTY → Payout = min(Bill, Max Coverage Amount)

Final Payout = Calculated Amount − Deductible (minimum ₹0)

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Integrations & Security

Razorpay — Premium Collection

- Customer pays policy premium online
- Backend creates Razorpay order in paise
- HMAC-SHA256 signature verified server-side
- Policy PDF and invoice auto-generated on success

RazorpayX — Claim Bank Payouts

- Approved claim triggers bank payout
- Contact and Fund Account created via API
- Bank transfer processed directly to customer
- Mock mode available for test environments

n8n Webhook — Email Automation

- Staff creation triggers welcome email
- Credentials sent via n8n webhook payload
- Email failure does not block staff creation
- Fully decoupled from backend transaction

Security Measures

- **JWT Authentication**
  Bearer token required on every API request
- **Spring RBAC**
  @PreAuthorize annotation on every controller
- **BCrypt Hashing**
  Passwords never stored in plain text
- **Input Validation**
  Regex validation for VIN format and registration number

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